

STRUTHIO ONE

Build manual · Windows 11 · every step in build order



From ordering the parts to playing: flash the full game, build it, test it.
Design locked · commit af4d4a2 · 2026-10-02

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How to read this manual

Do the parts in order. Each step has a picture, what to do, and a green ✓ **check**: don't go on until it's true. Orange boxes are the things that matter most.

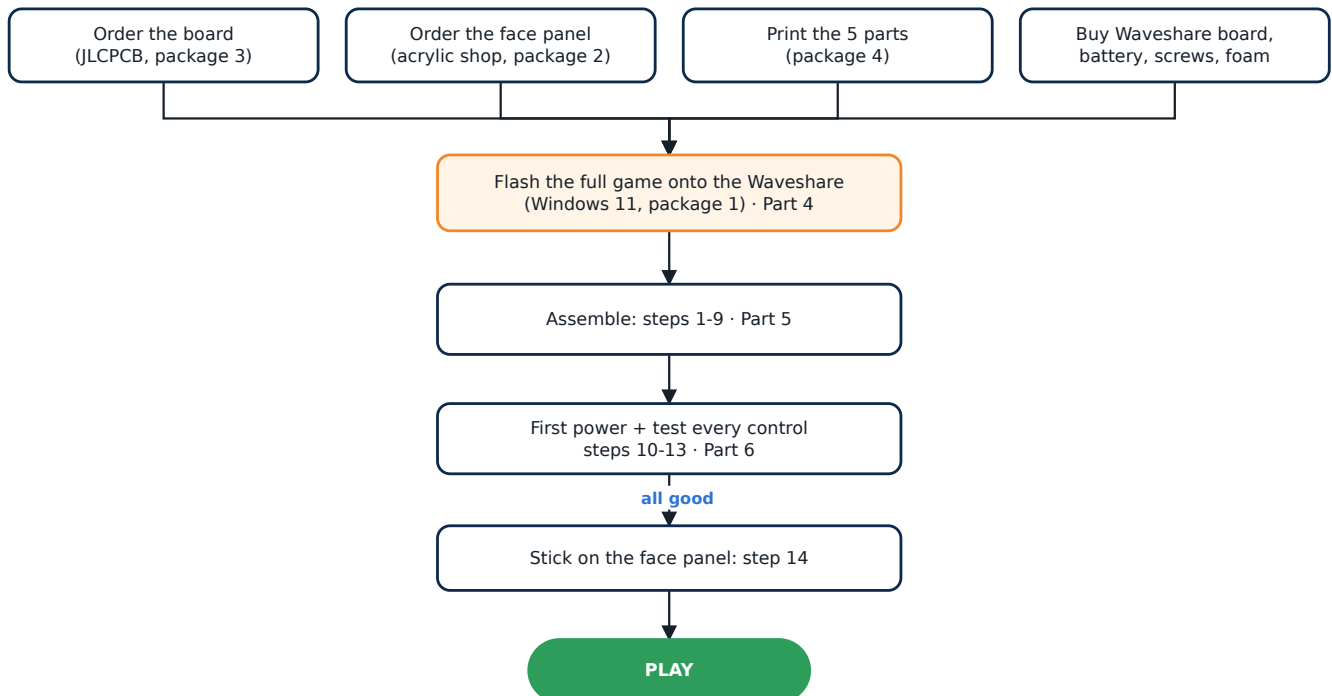
Pictures are labelled: **SCREENSHOT** real output of the real firmware and tools, **FROM THE 3D MODEL** rendered from the exact case and board design (the parts each step adds glow orange), **DIAGRAM** drawn from the design's own numbers, **ILLUSTRATION** a drawing of a Windows window to show where to click.

The six packages you need are referred to by number: **1** firmware flashing, **2** face panel print, **3** PCB (JLCPCB), **4** 3D print, **5** documents (this manual), plus the full backup.

The whole build at a glance

PART 1

Four things are made or bought at the same time; you flash the game as soon as the Waveshare board arrives, then build, test, and finish with the face panel.



The build, start to finish DIAGRAM



Everything that goes into it, front to back: face panel, front shell, buttons, Waveshare board, battery, ONE board, back shell FROM THE 3D MODEL

What you need

PART 2

Parts

✓ Item	Exactly what
☐ Waveshare ESP32-S3-Touch-LCD-3.5B	The bare board (not the "-C" cased version). Its 6 Ω 1 W speaker comes with it.
☐ LiPo battery	THOR-503450 (5 × 34 × 52 mm), 1000 mAh, ordered with a JST PH 2.0 mm 2-pin plug and an 80-100 mm lead.
☐ ONE board	From JLCPCB, assembled (Part 3).
☐ Face panel	Laser-cut, back-printed 1.0 mm acrylic (Part 3).
☐ 5 printed parts	Front shell, back shell, two wing buttons, rocker (Part 3).
☐ 2 screws	M2 × 8 pan head, self-tapping or machine.
☐ 1.75 mm filament	One 11 mm piece: the rocker's axle.
☐ Foam pad	3 mm soft foam, about 30 × 45 mm, self-adhesive on one side.
☐ Double-sided tape	Thin; for the speaker and the face panel (clear transfer tape is best for the panel).
☐ USB-C data cable	One that carries data, not a charge-only cable.
☐ Blue paint pen (optional)	To colour the engraved wings and dart.

Tools

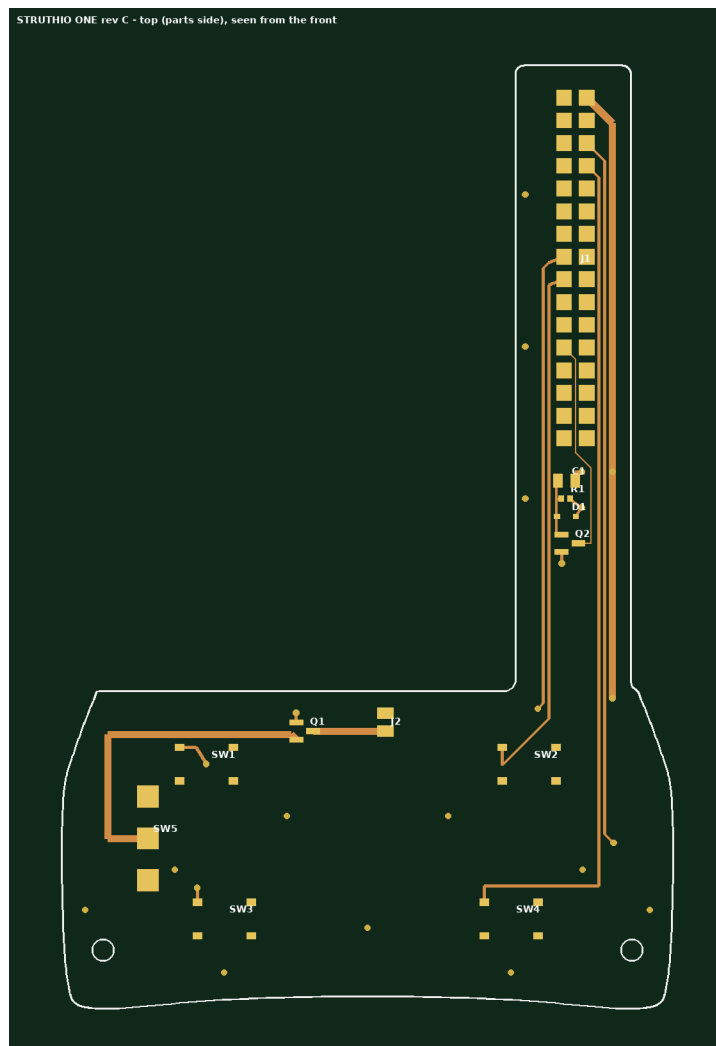
- ☐ A small Phillips screwdriver (PH0 or PH1) for the M2 screws
- ☐ Scissors, a craft knife
- ☐ A needle or pin (only if the battery plug needs its wires swapped)
- ☐ A paperclip (the side pin holes, for re-flashing later)
- ☐ A Windows 11 computer with internet access

Order and make the parts

PART 3

The ONE board (package 3, JLCPCB)

1. On jlcpcb.com upload `struthio_one_gerbbers.zip`: 2 layers, 1.6 mm, any colour.
2. Turn on **PCB Assembly**, top side, **Standard** PCBA (three parts are through-hole). Upload `BOM_JLCPCB.csv` and `CPL_JLCPCB.csv`.
3. In the placement preview, compare with the picture below: **J1** an ordinary header on the strip; **J2** opening toward the right edge; **SW5** knob off the left edge; **Q1**, **Q2** single leg as on the silkscreen; **D1** band toward the left. Rotate any part that differs in 90° steps.



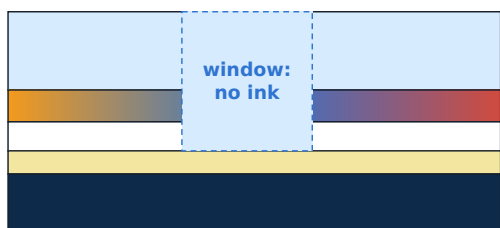
The board as ordered (top side, seen from the front of the handheld), with every part's place DIAGRAM

The face panel (package 2)

Send the shop the files and this sentence: **1.0 mm clear cast acrylic, cut to the DXF, reverse-printed (colour, then white) from the mirrored PNG, window left clear.** Cheaper: a clear cut panel and the art printed on sticker vinyl, stuck on its back.



The proof: cut lines magenta, the clear window blue



clear acrylic 1.0 mm (the front you touch)

colour print (on its back)

white underprint

double-sided tape (black areas only)

front shell face

What the shop makes, in layers (the window stays clear)

DIAGRAM

The printed parts (package 4)

PLA or PETG, 0.2 mm layers, 3 walls. One of each STL. The wing buttons and rocker print engraving-down: keep the first layer neat.

Side views, the bed at the bottom. No supports needed.



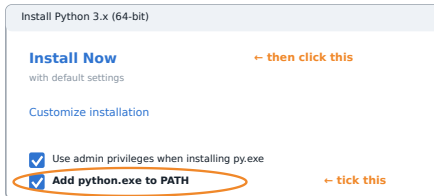
How each part sits on the bed

DIAGRAM

Flash the full game (Windows 11)

PART 4 · DO THIS AS SOON AS THE WAVESHARE BOARD ARRIVES, BEFORE YOU BUILD IT IN

This puts the whole game on the board: the program, the art pack and the soundtrack. You need nothing but Python and one command. The board stays on its own on the desk, connected by USB-C.



The Python installer ILLUSTRATION

4.1 Install Python

Download Python 3 from **python.org/downloads** and run it. Tick **"Add python.exe to PATH"**, then click **Install Now**.

✓ Python installed.

4.2 Install the flashing tool

Right-click the **Start** button → **Terminal**. Type, then Enter:

```
PS> py -m pip install "esptool>=5"
```

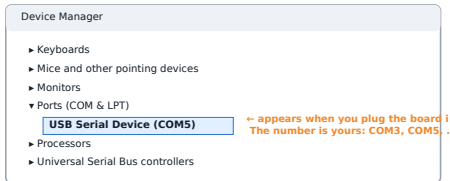
✓ The last line says "Successfully installed esptool-5..." (or "Requirement already satisfied").

4.3 Unzip the firmware package

Right-click `STRUTHIO_ONE_1_Firmware_Flashing.zip` → **Extract All...** → type `C:\` → Extract. You now have `C:\struthio`:

```
C:\struthio\  
├─ prebuilt\           ← the game, ready to flash (you run the command here)  
│   ├── bootloader.bin, partition-table.bin, struthio.bin  
│   └─ flash_args.txt  (addresses and settings for esptool)  
├─ handheld\build\assets\ ← the art pack and the soundtrack  
├─ manual\  
├─ STRUTHIO.bat        (the menu, for building it yourself)  
└─ README_FIRST.txt
```

Not inside OneDrive, and no spaces in the path: `C:\struthio` exactly.



Device Manager ILLUSTRATION

4.4 Plug the board in, find its port

Connect the WaveShare to the PC with the USB-C **data** cable. Right-click **Start** → **Device Manager** → **Ports (COM & LPT)**. The entry that appears when you plug it in is the board. Remember its number, e.g. **COM5**.

Nothing appears? Try another cable (many only charge) and another USB socket.

✓ You know your board's COM number.

4.5 Go to the prebuilt folder

In the Terminal:

```
PS> cd C:\struthio\prebuilt
```

4.6 Flash

Type this, with **your** COM number, then Enter. It takes a few minutes; don't unplug.

```
PS C:\struthio\prebuilt> py -m esptool --chip esp32s3 -p COM5 -b 460800 write-flash @flash_args.txt
```

✓ Five parts, each ending with "Hash of data verified." (screenshot below).

```
Terminal — ~/struthio/prebuilt (the emulator bench)

maker@bench:~/struthio/prebuilt$ python -m esptool --chip esp32s3 -p socket://localhost:5555 -b 460...
esptool v5.4.0

Connected to ESP32-S3 on socket://localhost:5555: 1
Chip type: ESP32-S3 (QFN56) (revision v0.3)

Stub flasher running.

Writing 'bootloader.bin' at 0x00000000...

Wrote 20848 bytes (13355 compressed) at 0x00000000 in 0.8 seconds (216.0 kbit/s).

Hash of data verified. 2

Writing 'struthio.bin' at 0x00010000...

Wrote 608496 bytes (394779 compressed) at 0x00010000 in 19.3 seconds (252.7 kbit/s).

Hash of data verified. 2

Writing '../handheld/build/assets/struthio.pak' at 0x00410000...

Wrote 8661084 bytes (1526098 compressed) at 0x00410000 in 85.2 seconds (813.3 kbit/s).

Hash of data verified. 2

Writing '../handheld/build/assets/struthio_music.ima' at 0x00d10000...

Wrote 1920020 bytes (1745564 compressed) at 0x00d10000 in 85.1 seconds (180.4 kbit/s).

Hash of data verified. 2

Hard resetting via RTS pin...
```

The real output of that command. **1** connected; **2** each part written and checked. Captured against Espressif's ESP32-S3 emulator, so its port reads `socket://localhost:5555` where yours says **COM5**, its prompt is the bench computer's, and its command shows `python` where you type `py`; lines left out are marked `...` [SCREENSHOT](#)



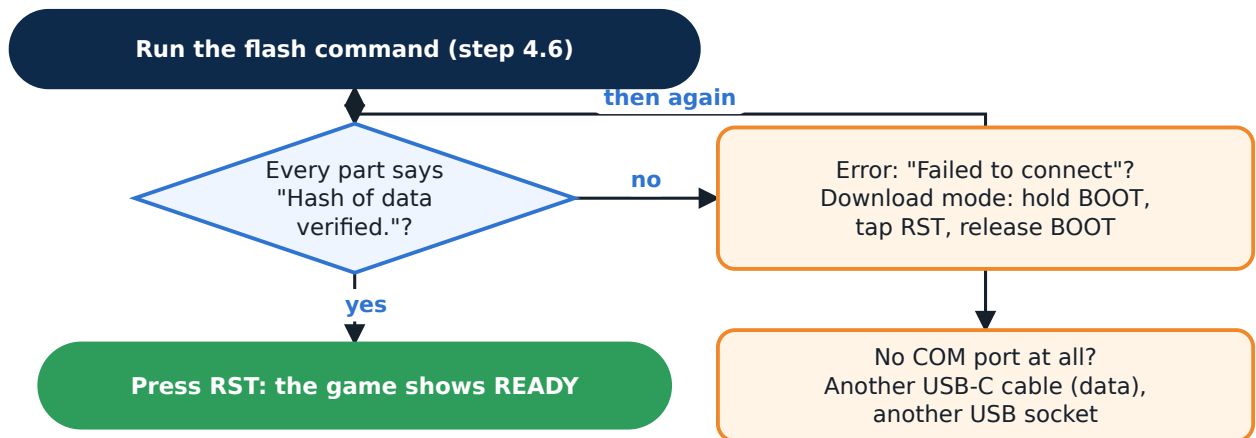
What the board shows: the game waiting at **READY** [SCREENSHOT](#)

4.7 Start the game

Press the board's **RST** button once. The game appears with **READY** in the middle. With no buttons yet, it waits there: that's correct.

- ✓ **READY** on the screen. Unplug the board: it's ready to build in.

If flashing doesn't work



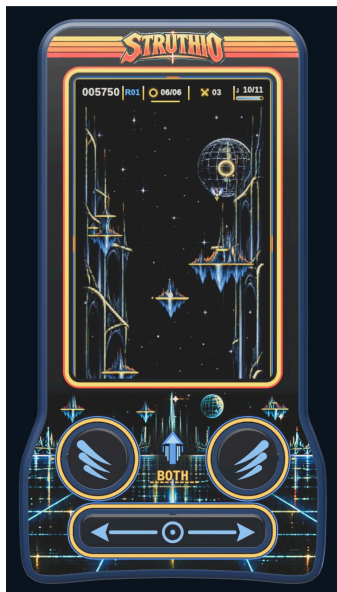
Flashing problems DIAGRAM

Download mode (only if it won't connect): hold the board's **BOOT** button, press and release **RST**, release **BOOT**, then run 4.6 again. The COM number can change in download mode: look again in Device Manager. After a later re-flash through the finished case, the same buttons are behind the pin holes on the left side (Part 8).

Prefer clicking to typing? Espressif's browser flasher espressif.github.io/esptool-js (Chrome or Edge) does the same: add `prebuilt\bootloader.bin` at `0x0`, `partition-table.bin` at `0x8000`, `struthio.bin` at `0x10000`, `handheld\build\assets\struthio.pak` at `0x410000`, `struthio_music.ima` at `0xd10000`.

Assemble

PART 5 · STEPS 1-9 · WORK ON A SOFT CLOTH, NOTHING NEEDS FORCE



The engraved wings and dart, filled
blue FROM THE 3D MODEL

1 Paint (optional)

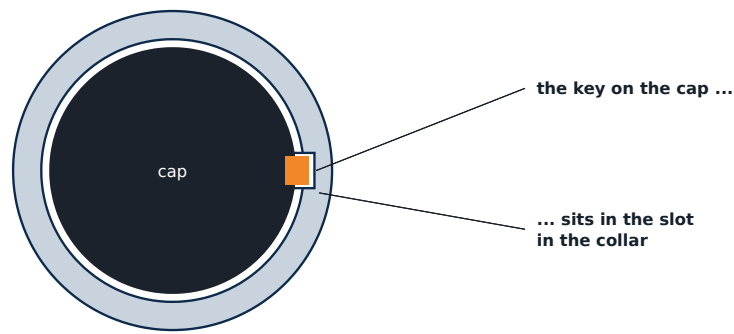
Fill the engraved wings and the dart with the blue paint pen. Wipe the top faces clean, let it dry.



2 Fit the wing buttons

From the **inside** of the front shell, push each round cap into its collar with its **key** (the small tab on its side) in the slot on the collar's outer side.

- ✓ Both caps are in, keys in their slots, wings upright from the front. Each one springs back when pressed.



Why it only goes in one way: a wing button seen from the inside of the front shell; the key points away from the middle

DIAGRAM



3 Fit the rocker

Push the rocker into its collar from the inside. Cut an **11 mm** piece of 1.75 mm filament and push it through the hole in the collar's top wall until it stops.

✓ The rocker tips freely both ways and springs back.



4 Lay in the Waveshare

Screen **down**, **USB-C at the top** (it lines up with the opening), its PWR/RST/BOOT buttons toward the pin holes on the left. It drops between the ribs.

✓ It lies flat and doesn't slide.

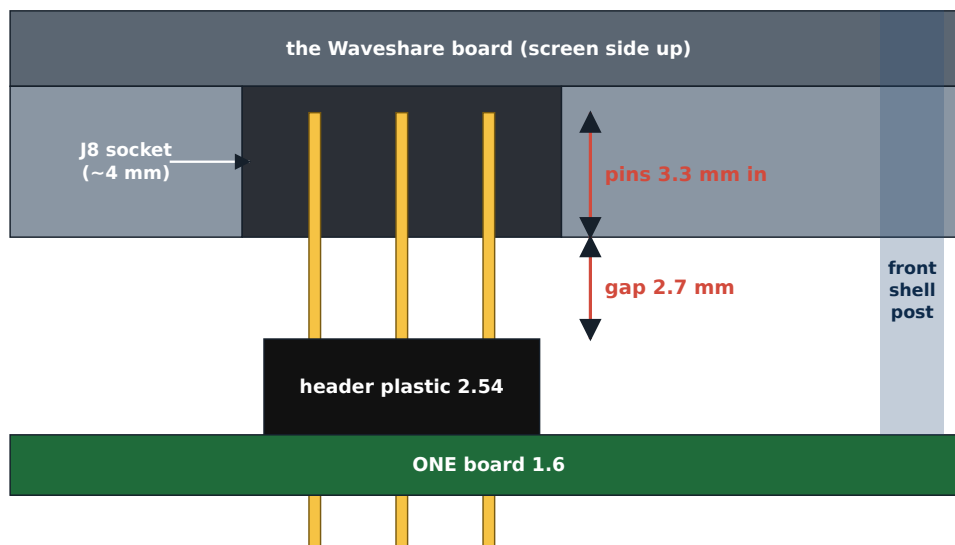


5 Plug on the ONE board

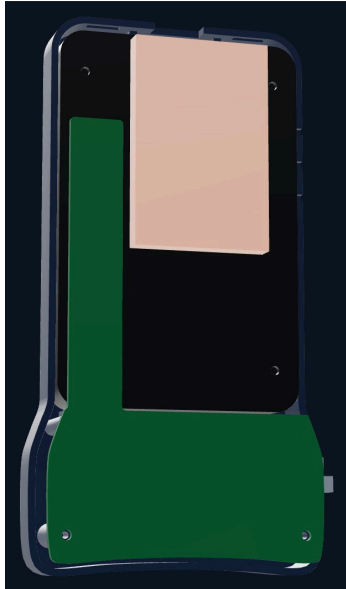
Set the slide switch **down (OFF)**. Hold the board over the Waveshare, header over the J8 socket, lower it straight and press evenly until it **rests on the four posts**.

It stops on the posts by itself. Don't press harder: the pins are meant to go only 3.3 mm in.

✓ The ONE board sits flat on all four posts.



How deep the header goes (side view, front at the top). The ONE board rests on the front shell's posts, and that sets the depth: the pins go 3.3 mm in and the plastic never touches the socket DIAGRAM

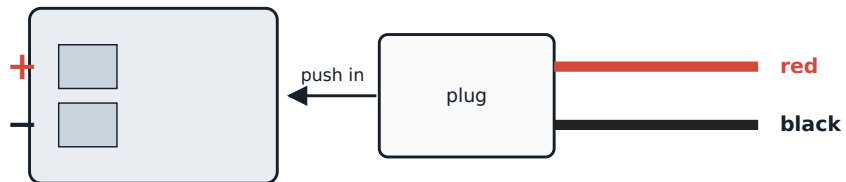


6 Battery

Check the plug against the + printed beside J2: **red to +**. Plug it into J2 from the right. Lay the cell on the Waveshare's back left of the strip, **lead end down**, the lead flat in the gap down to the plug.

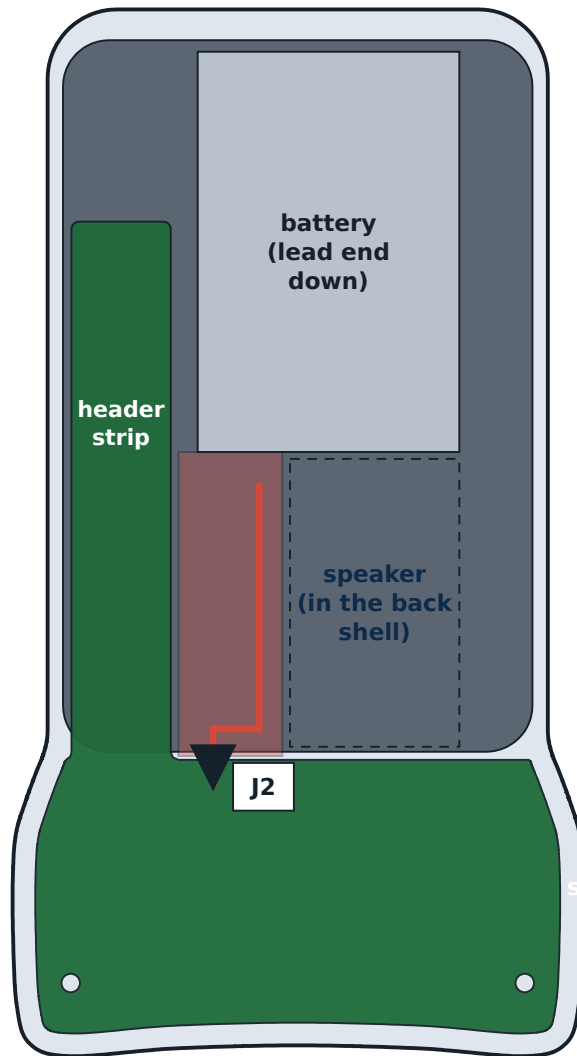
✓ Red wire on +, lead lying flat, nothing pinched.

J2 on the ONE board (white socket)

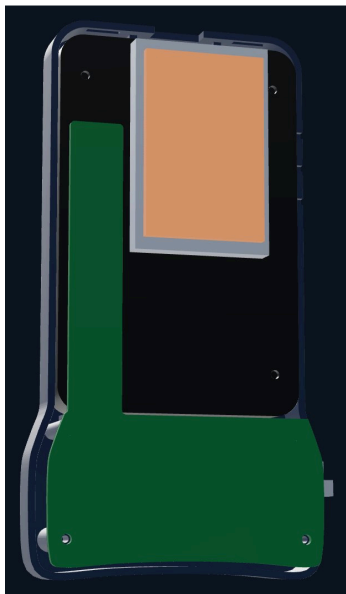


RED wire on the + side

Battery plug polarity. Plugged the wrong way round it does nothing (Q1 blocks it): swap the two crimp contacts (lift each latch with a needle) and plug in again [DIAGRAM](#)



Where things lie, seen from the back while you build (USB-C at the top). Red: the battery lead, flat on the Waveshare's back between the cell and the strip, down to J2 DIAGRAM



7 Foam pad

Stick the 3 mm foam pad on top of the battery. It presses the battery and the Waveshare forward when the case closes.



8 Speaker into the back shell

Tape on the speaker's back, stick it in the bay inside the back shell, **face toward the grille**. Plug its lead into **J9** on the Waveshare (small 2-pin socket at its left edge, beside BOOT). Keep the back shell right beside the case: the lead is short.

✓ Speaker in its bay, plug in J9.



9 Close the case

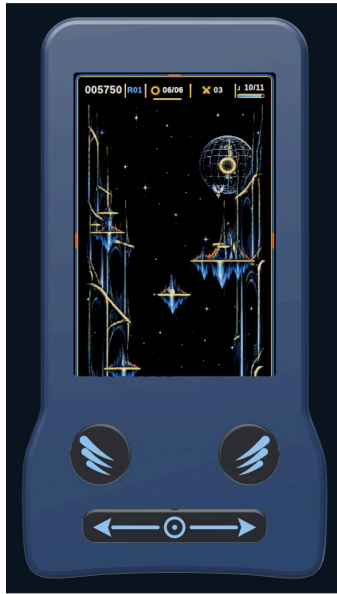
Put the back shell's two top hooks into the slots inside the front shell's top edge and swing it down. The power knob slides into its slot, three pegs drop into the Waveshare's holes. Fit the two M2 × 8 screws at the bottom, **snug, not tight**.

If it doesn't close by itself, open it and look: a lead or the foam is in the way. Never force it.

✓ Closed all round, no gap at the seam.

First power and test

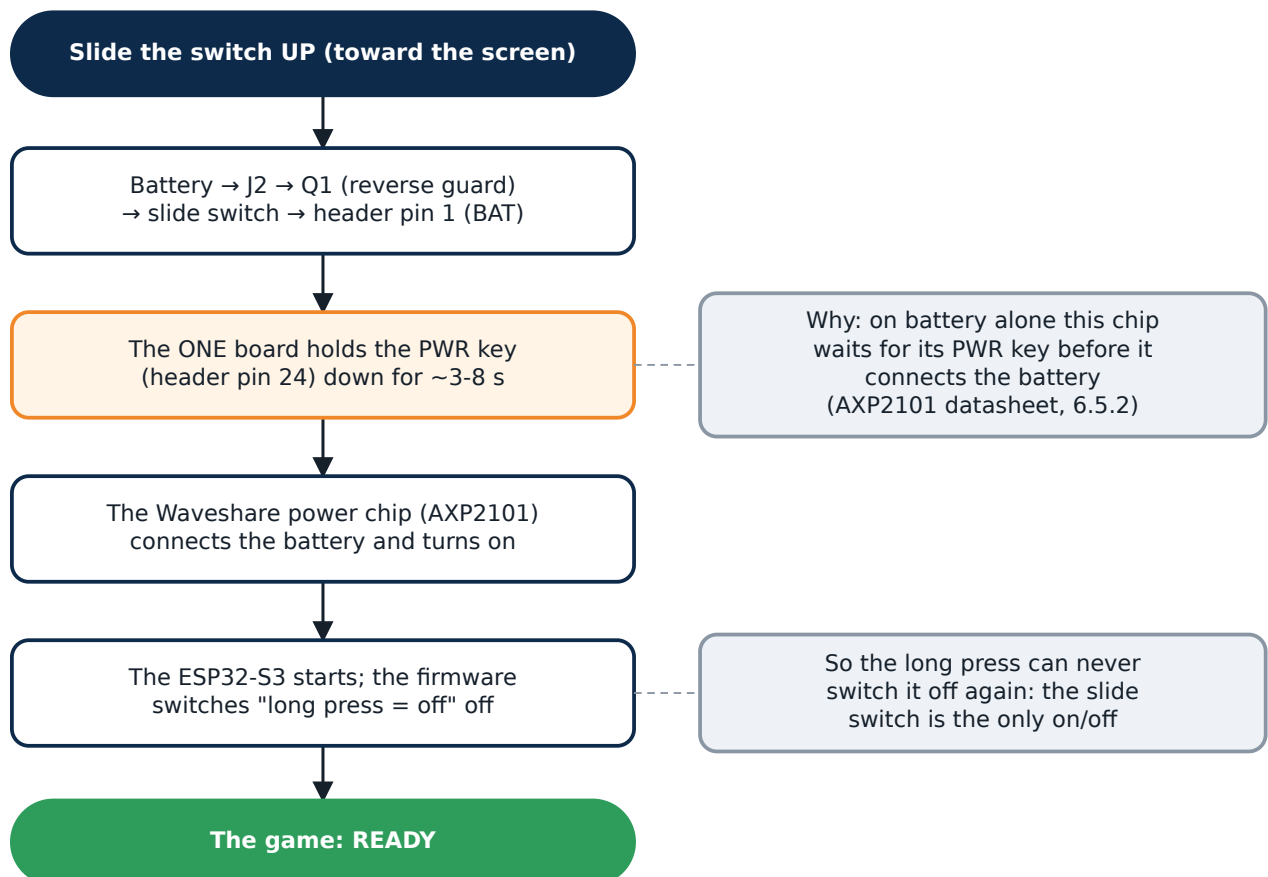
PART 6 · STEPS 10-13



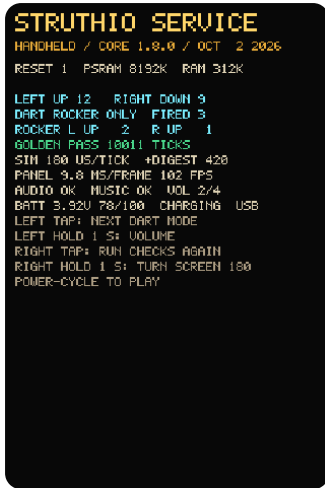
10 Switch on

Slide the switch **up**. Within about 3-8 s the screen lights and the game starts. Slide it down: off at once.

✓ The game appears. (Nothing? Part 9.)



What happens when you slide the switch up **DIAGRAM**



The service screen (example values)

SCREENSHOT

11 Test every control

Hold **both wing buttons**, slide the switch up, keep holding until the service screen appears. Press each wing button and each rocker end: its counter goes up by one per press, and none counts by itself. **BATT** shows the battery (**BATT NONE:** the cell isn't connected).

Switch off to leave.

- ✓ LEFT, RIGHT, ROCKER L and ROCKER R each count one per press; BATT shows a voltage.

12 Picture the wrong way up?

In service mode, **hold RIGHT for 1 s**. It's saved. Switch off and on to play.

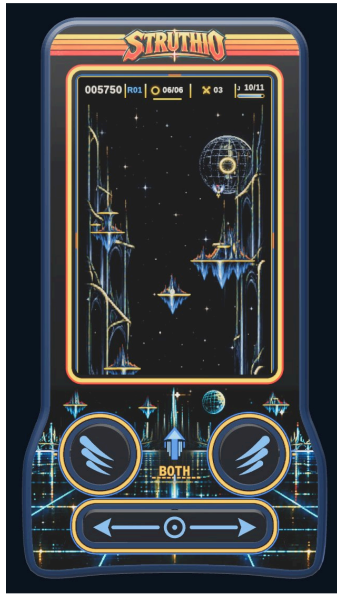
13 Charge

Plug USB-C in **with the switch ON**. With the switch OFF the battery is disconnected: USB still runs the game but doesn't charge.

- ✓ In service mode BATT says CHARGING and USB.

Fit the face panel

PART 7 · STEP 14, LAST



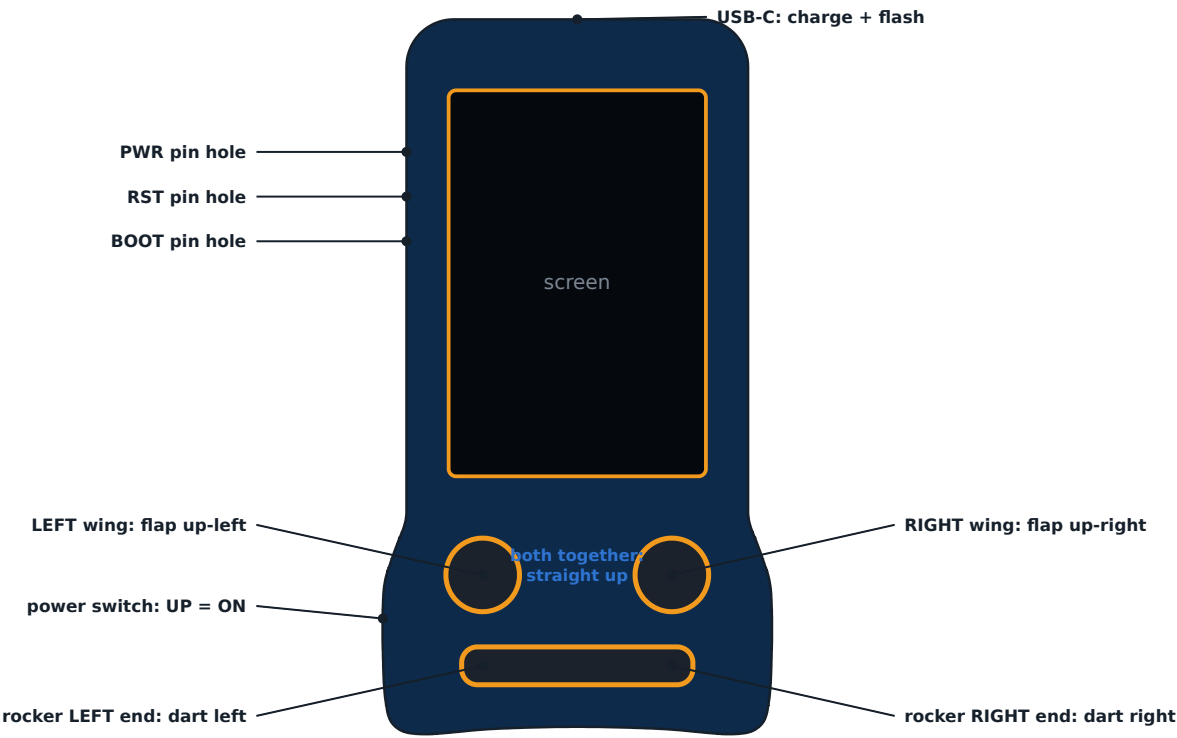
14 Face panel

Peel the protective film off the back. Tape on the **black areas only**. Line the panel up inside the rim, the wells over the buttons, and press it down from the middle outward. Peel the front film.

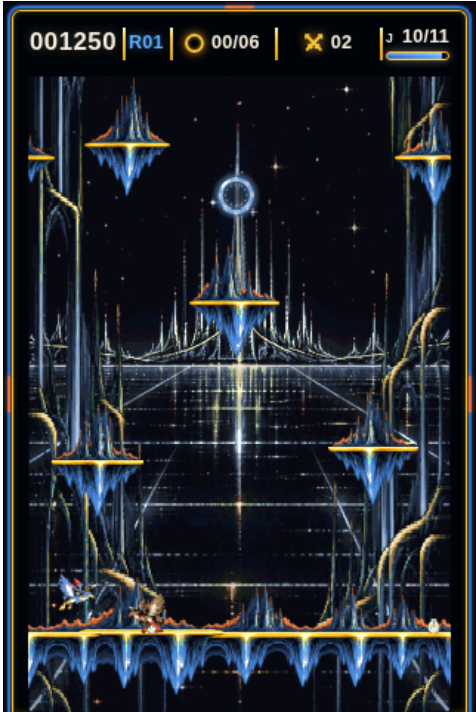
- ✓ The panel sits flat inside the rim; every button moves freely in its well.

How to play

PART 8



The controls, and what is on the sides DIAGRAM



The game running SCREENSHOT

The controls

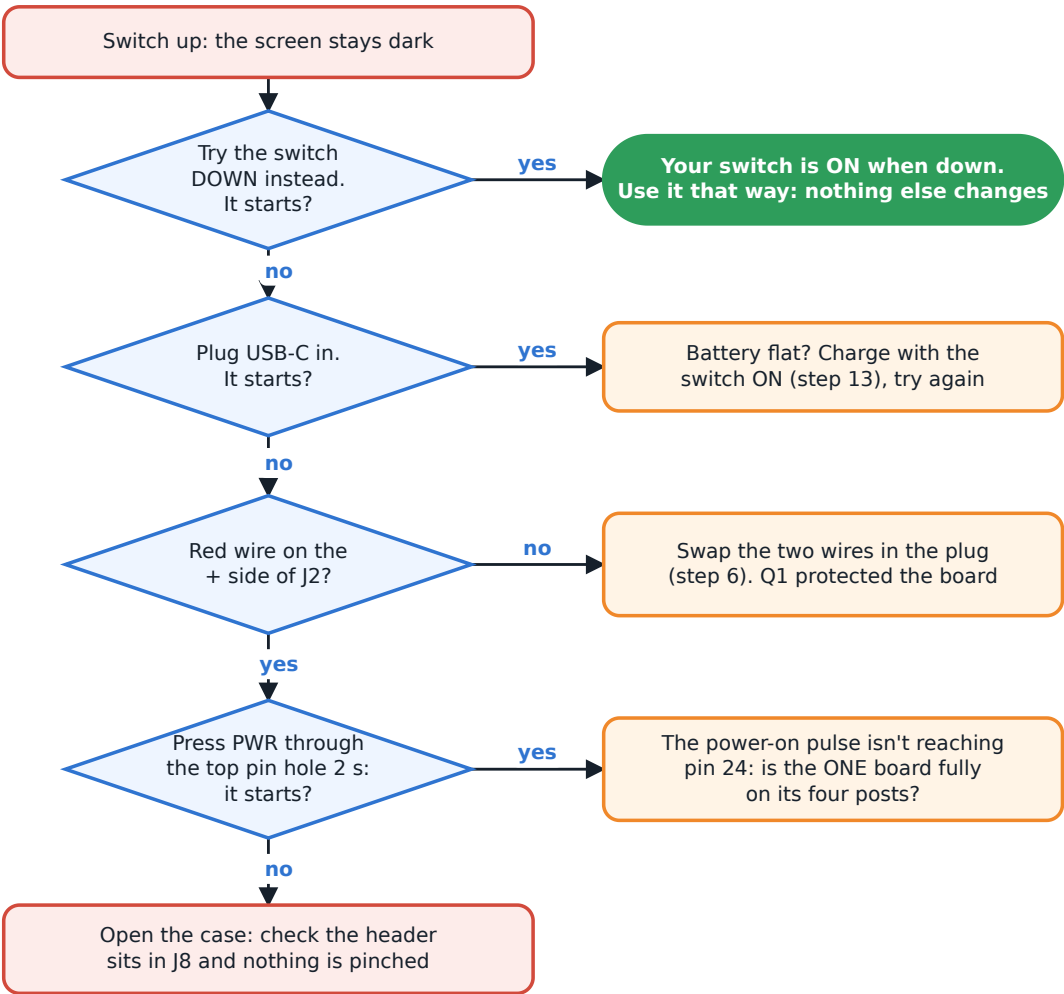
Press	Does
LEFT wing	flap up and to the left
RIGHT wing	flap up and to the right
Both together	flap straight up
Hold a wing	steer
Rocker left / right end	aim-assisted dart, that way
Both held 1 s after GAME OVER	new run
Both held while switching on	service mode

Updating the game later

Same as Part 4 through the USB-C in the top edge; the switch can be on or off. Download mode through the case: a paperclip in the **BOOT** hole, tap **RST**, release.

If something is wrong

PART 9

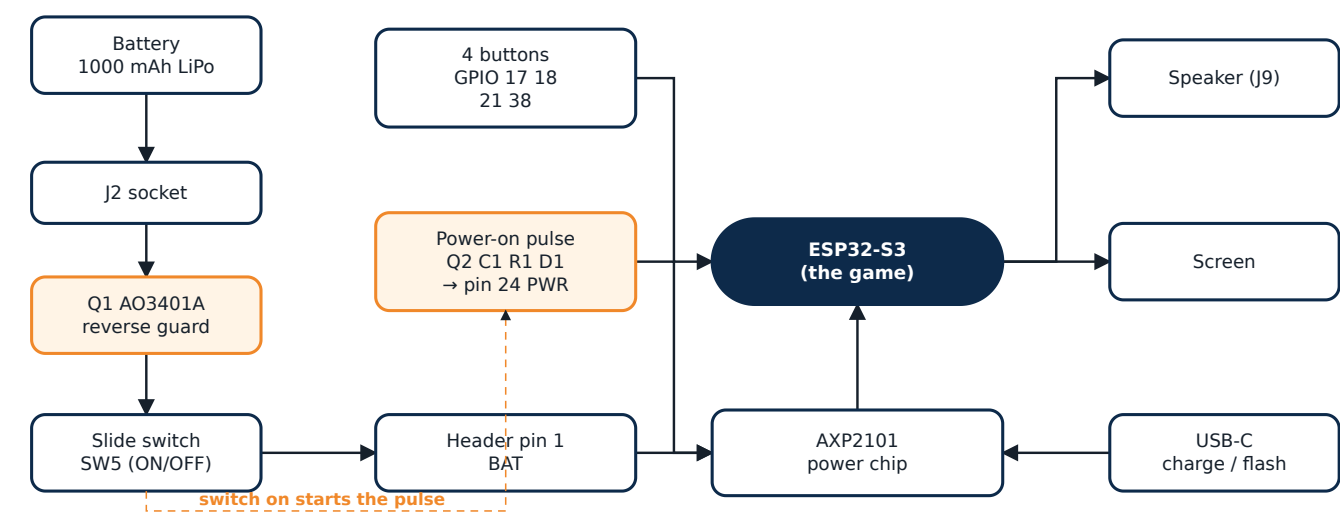


The screen stays dark when you switch on DIAGRAM

What you see	What to do
A button counts by itself (service mode)	Its stem touches the switch: take the cap out, sand 0.2 mm off the stem tip.
A button doesn't count	Is the ONE board on all four posts? Does the cap move freely?
No sound	Speaker plug in J9? Speaker face against the grille?
Back shell won't close	A lead or the foam is in the way. Knob in its slot? Never force it.
Picture upside down	Service mode, hold RIGHT 1 s.
Flashing fails	Part 4, "If flashing doesn't work".

How it works

PART 10



The electrics: what the ONE board does DIAGRAM



The 23 mm stack, front to back DIAGRAM

The ONE board

Part	Job
J1 header	into the Waveshare J8: BAT, GND, 4 buttons, PWR
SW1-SW4	the wing buttons and rocker ends
SW5	the slide switch: a real battery cut
J2	the battery plug
Q1	reverse-battery guard

Q2, C1, R1, D1	power-on pulse on PWR (pin 24)
----------------	--------------------------------

Header pins used

1 BAT · 3, 4, 29, 30 GND · 5 GPIO21 (rocker L) · 7 GPIO38 (rocker R) · 16 GPIO17 (left wing) · 18 GPIO18 (right wing) · 24 PWR

Things to confirm on the first build

What	Designed for	If different
J8 socket depth	~4 mm, pins 3.3 mm in	pins must hold; a 0.3 mm shim under the posts changes the depth
Waveshare hole size	≥ 2.2 mm (pegs 1.8)	sand the pegs
Speaker size, lead	bay 22 × 37 × 7.5 mm, lead ≥ 60 mm	any box that fits, with tape
Switch ON direction	up	use it the other way
USB-C height	opening 12.8 × 7.8 mm	file it wider
Acrylic thickness	1.0 ± 0.1 mm	the panel may sit a hair proud

STRUTHIO ONE build manual · commit af4d4a2 · 2026-10-02. Made from the design files by tools/manual/one_manual.py.